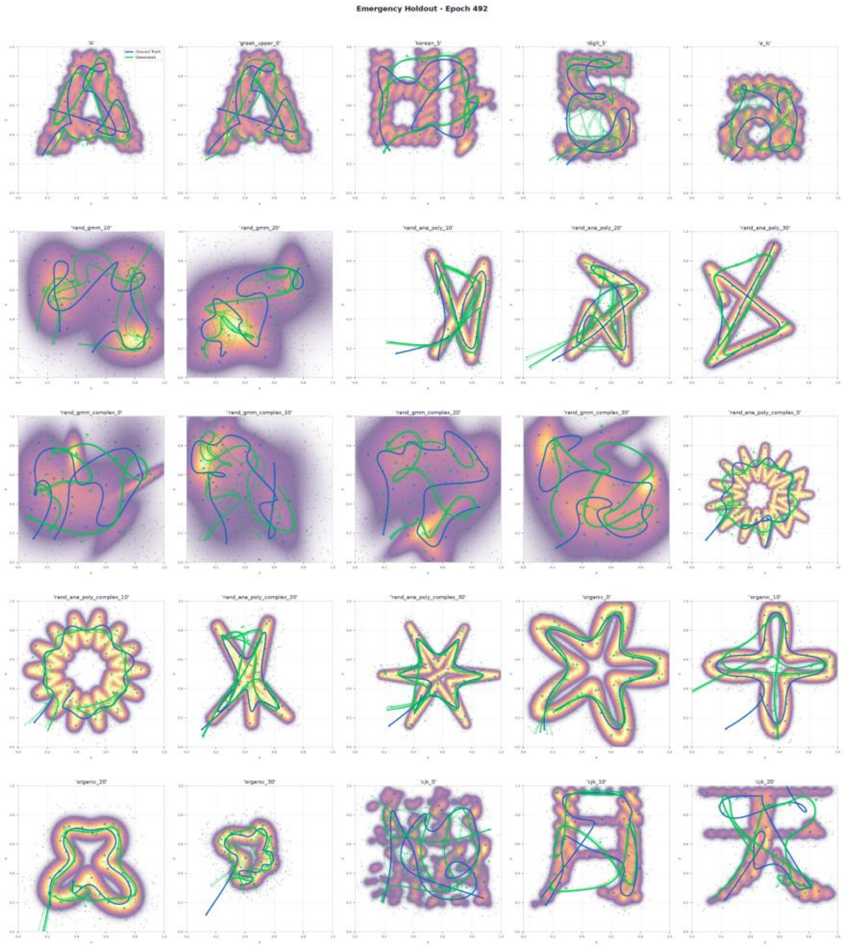
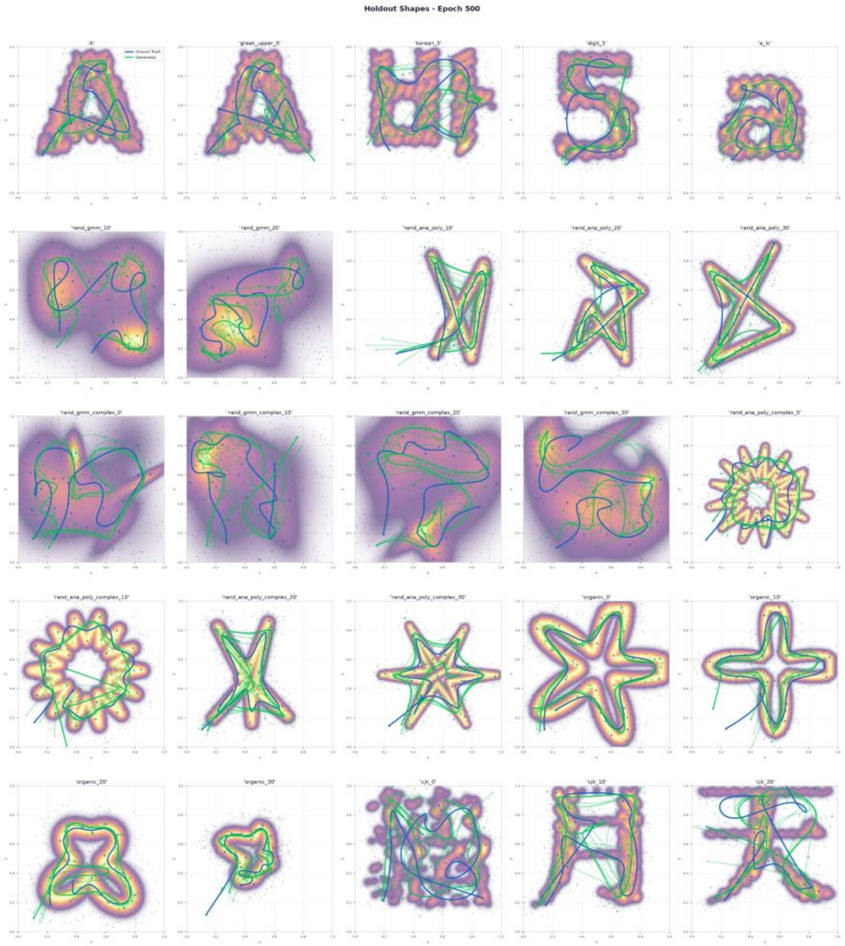
Meeting 27.08

Ergodic loss terms

	Ergodic error (Fourrier)
CFM	4,651
λ = 2	4,254
λ = 25	3,158
λ = 100	2,460
λ = 200	2,245
λ = 400	2,140
SVGD-Solver (“Groundtruth” from db)	2,709

Ergodic loss terms

λ = 2	λ = 400
	

Belief as Target Density

- Combine two target distributions each represented as GP:
  - $\mu(x)$:= known target distribution
  - $\sigma(x)$:= uncertainty distribution
  - $\Phi(x) = \mu(x) + \kappa \cdot \sigma(x)$

- Different planning approaches:
  - Ergodic planning on $\Phi(x)$
  - First step exploration of $\sigma(x)$, update $\mu(x)$ than exploitation of $\mu(x)$
  - Or compromiss:
    - Ergodic planning on $\Phi(x)$
    - Execute part of path, update $\Phi(x)$ replan and continue
    - ➔ Problem: no startpiontconditioning
    - Trained on new databank with random startpoints and included starpointconditioning via FILM architecture (same as time conditioning)

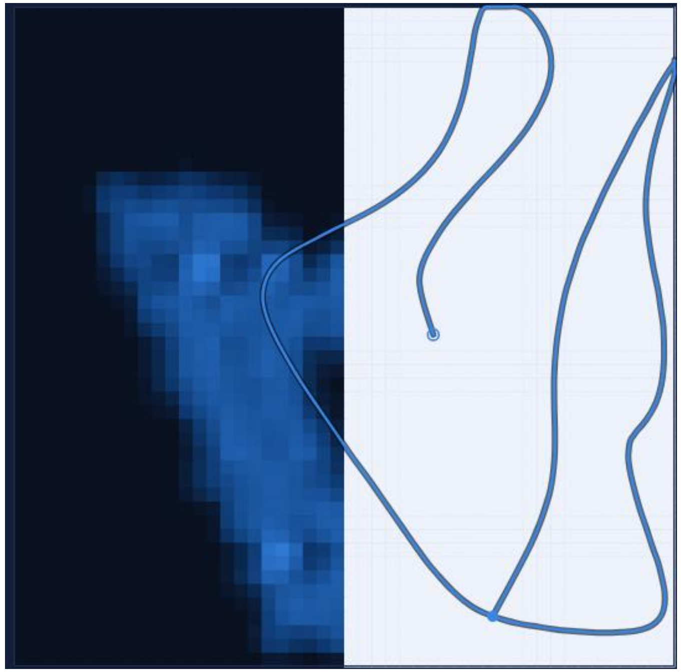
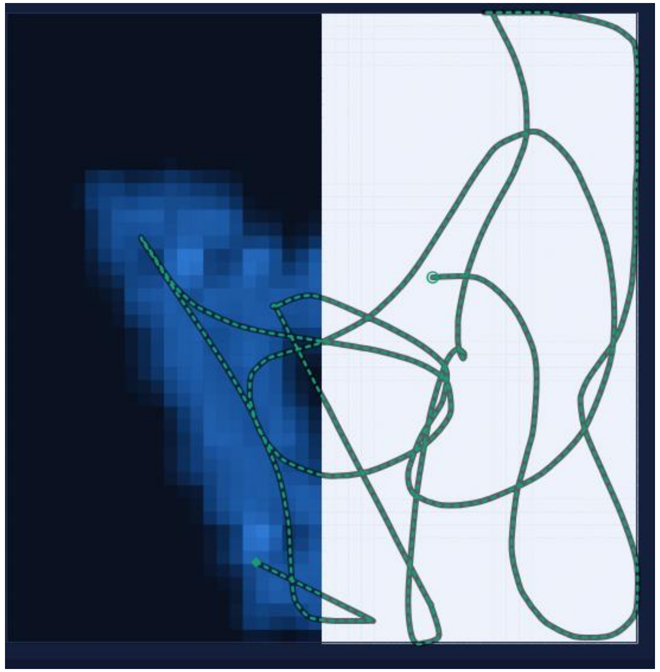
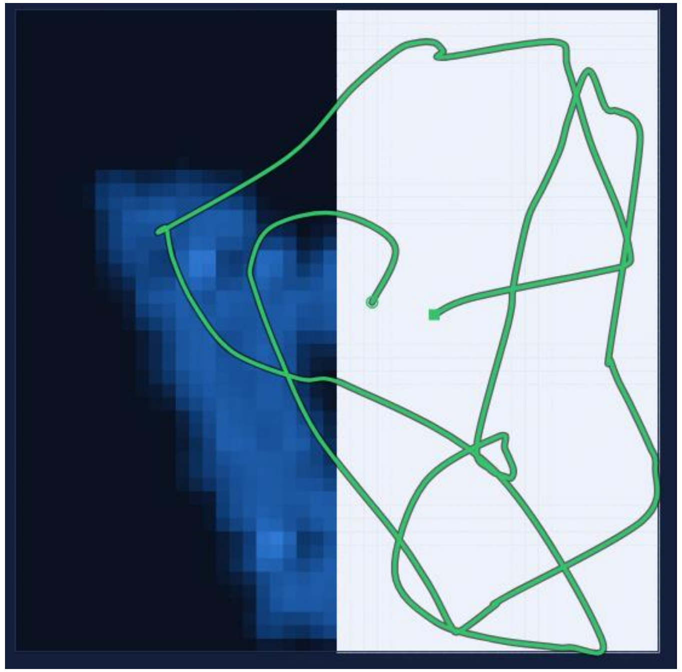
Belief as Target Density

- Test setup:
  - Exploration kernel length of 0,08 around exploration path
- Different starts:
  - 60 Randomly measured points on map
  - Half-half known/unknown
  - Different quadrants known/unknown
  - Center hole is unknown

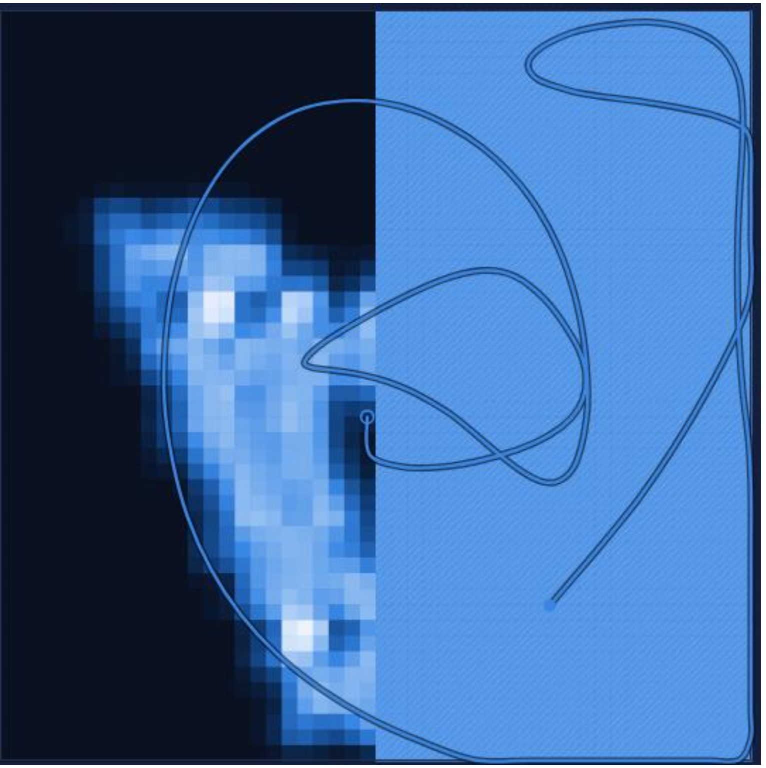
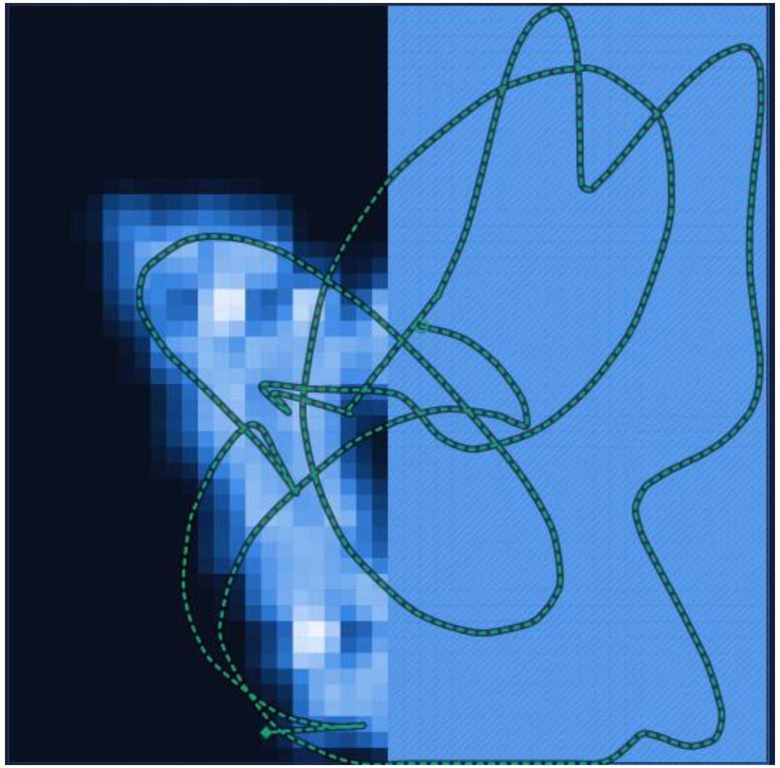
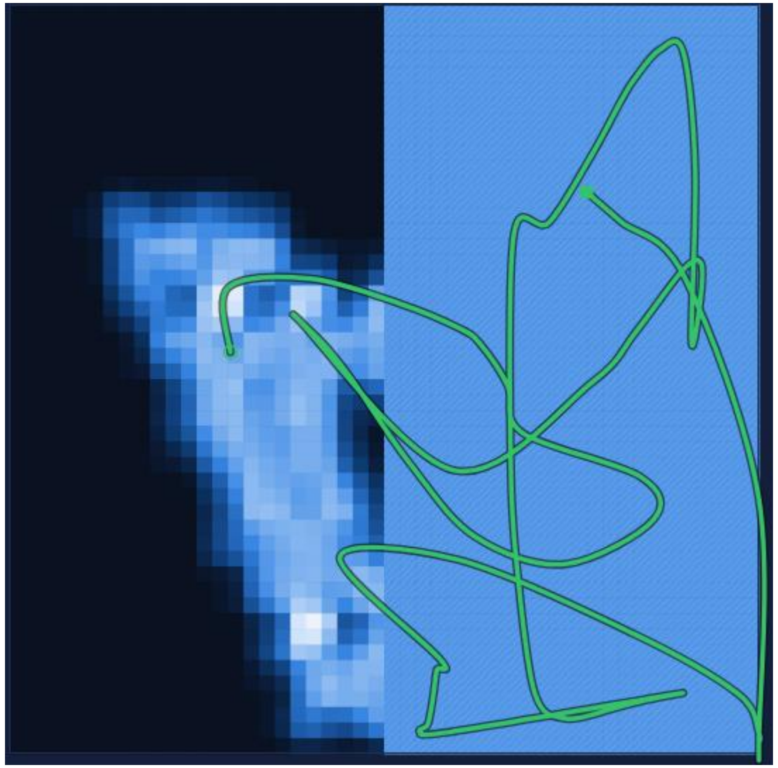
Belief as Target Density

- Test different target density modelation
  - Mass equalization: $\Phi(x) = (1-w)\ \mu(x) + w \cdot \sigma(x)$
    - W as time for exploration weight vs expoitation (w=0.5 means equal exploration and exploitation)
- Fisher-Information idea from paper “Ergodic Exploration of Distributed Information”
  - Idea: Where will another measurement deliver the most information gain?
  - Information gain or change in information density: $||\nabla\mu(x)||^2$ plus where is high information density for exploration: $\hat{\mu}(x)$
  - Math: $\Phi(x) = \hat{\mu}(x) + \kappa||\nabla\mu(x)||^2$

Ergodic planning on UCB

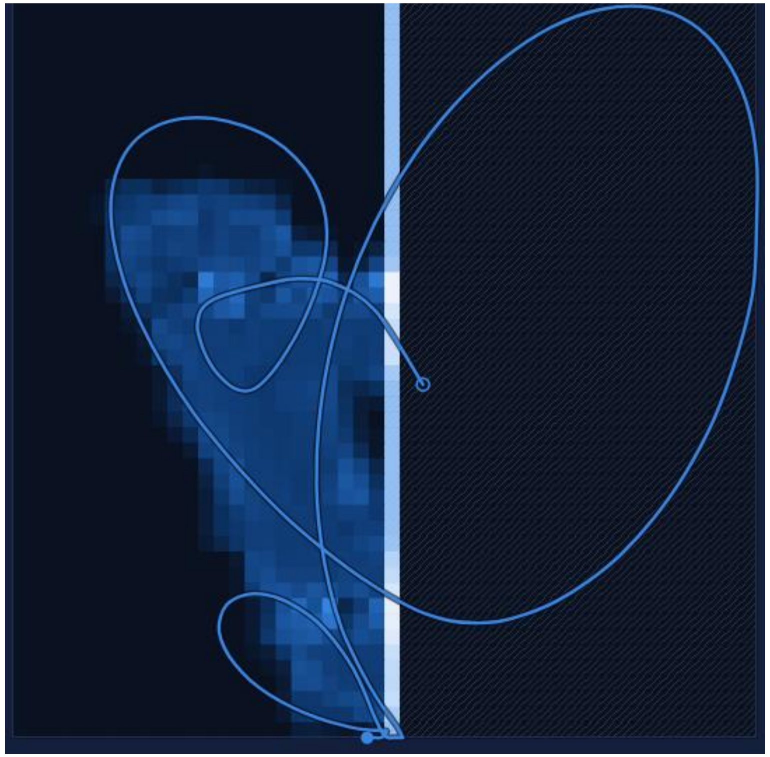
One time planning	Two stage planning	Constant replanning
		

Ergodic planning on UCB with Mass equalization

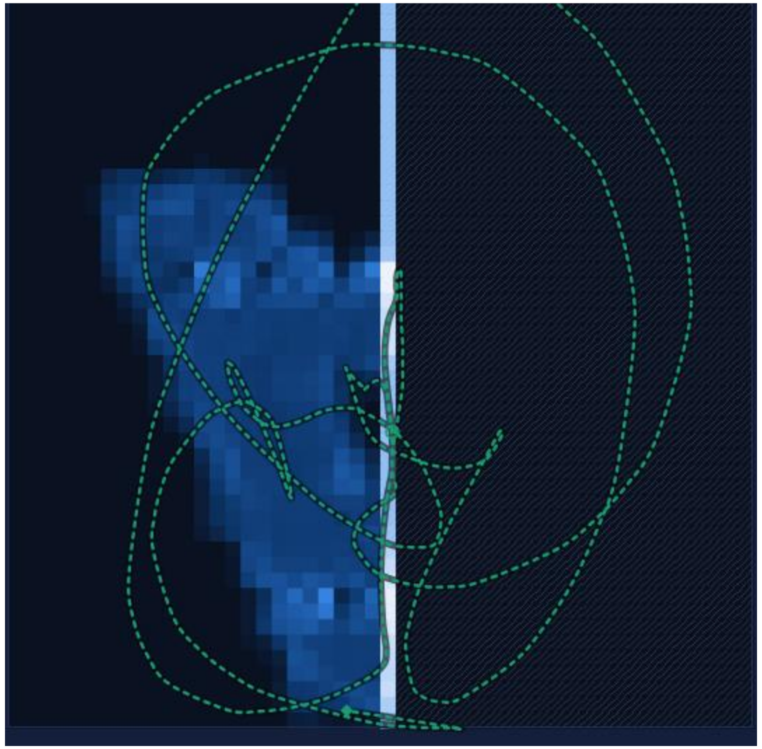
One time planning	Two stage planning	Constant replanning
		

Ergodic planning on Fisher-Information

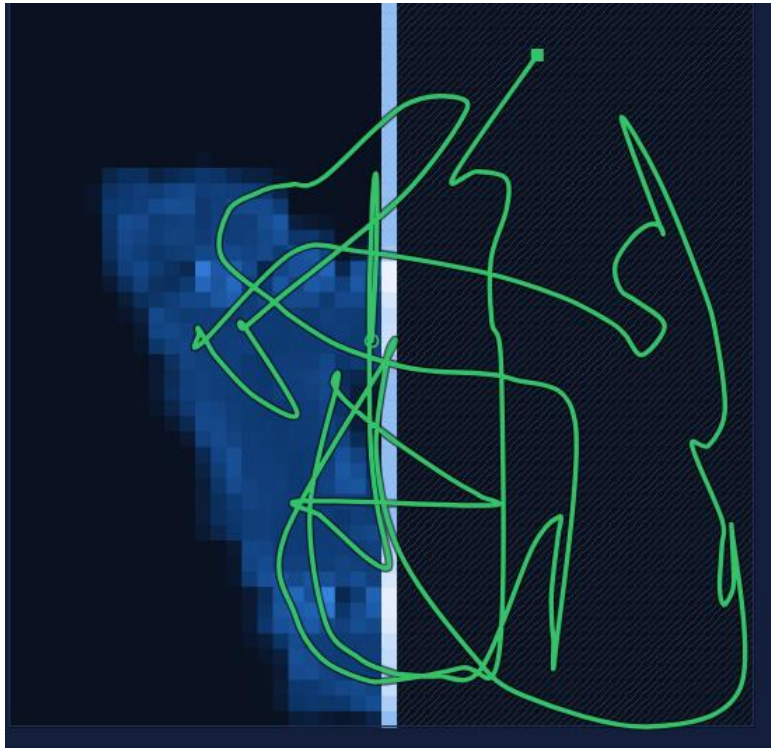
One time planning



Two stage planning



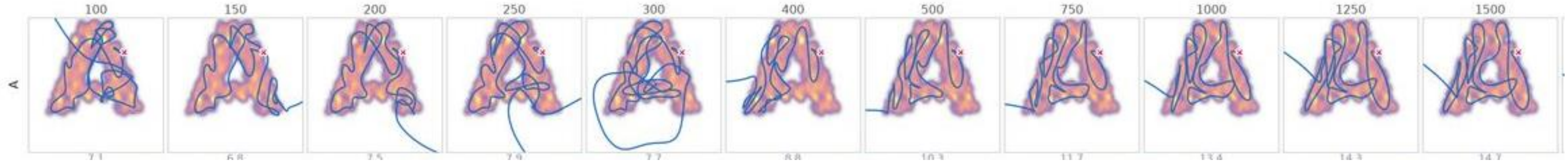
Constant replanning



Problem: length

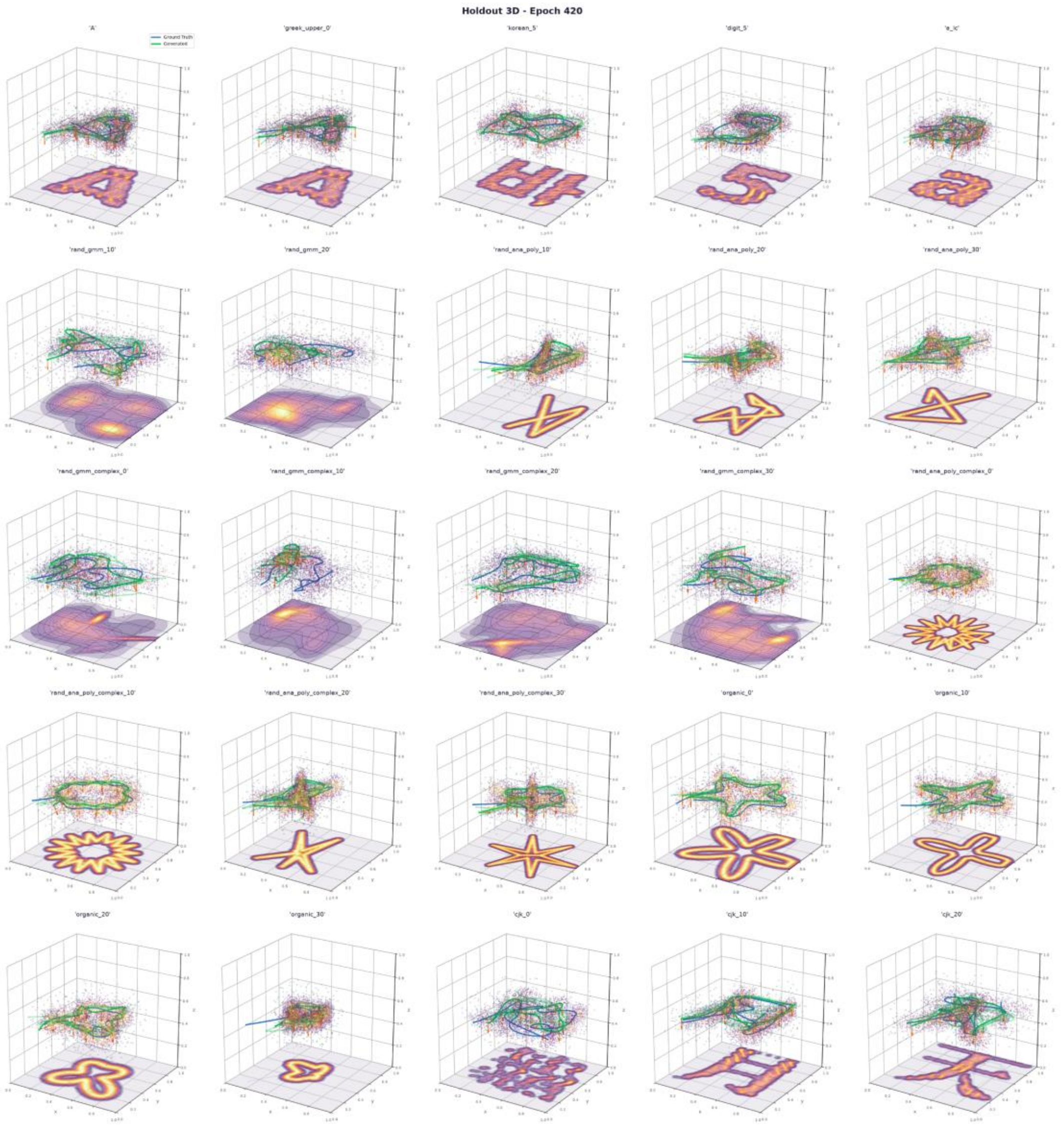
- The longer the trajectory is, the better the result, hard to define stop or limit length
- especially for the constant replanning
- ➔ idea if it is possible to condition on length, each replanning step can reduce length

- Databank with different iteration numbers for different lengths, no successfull inference yet



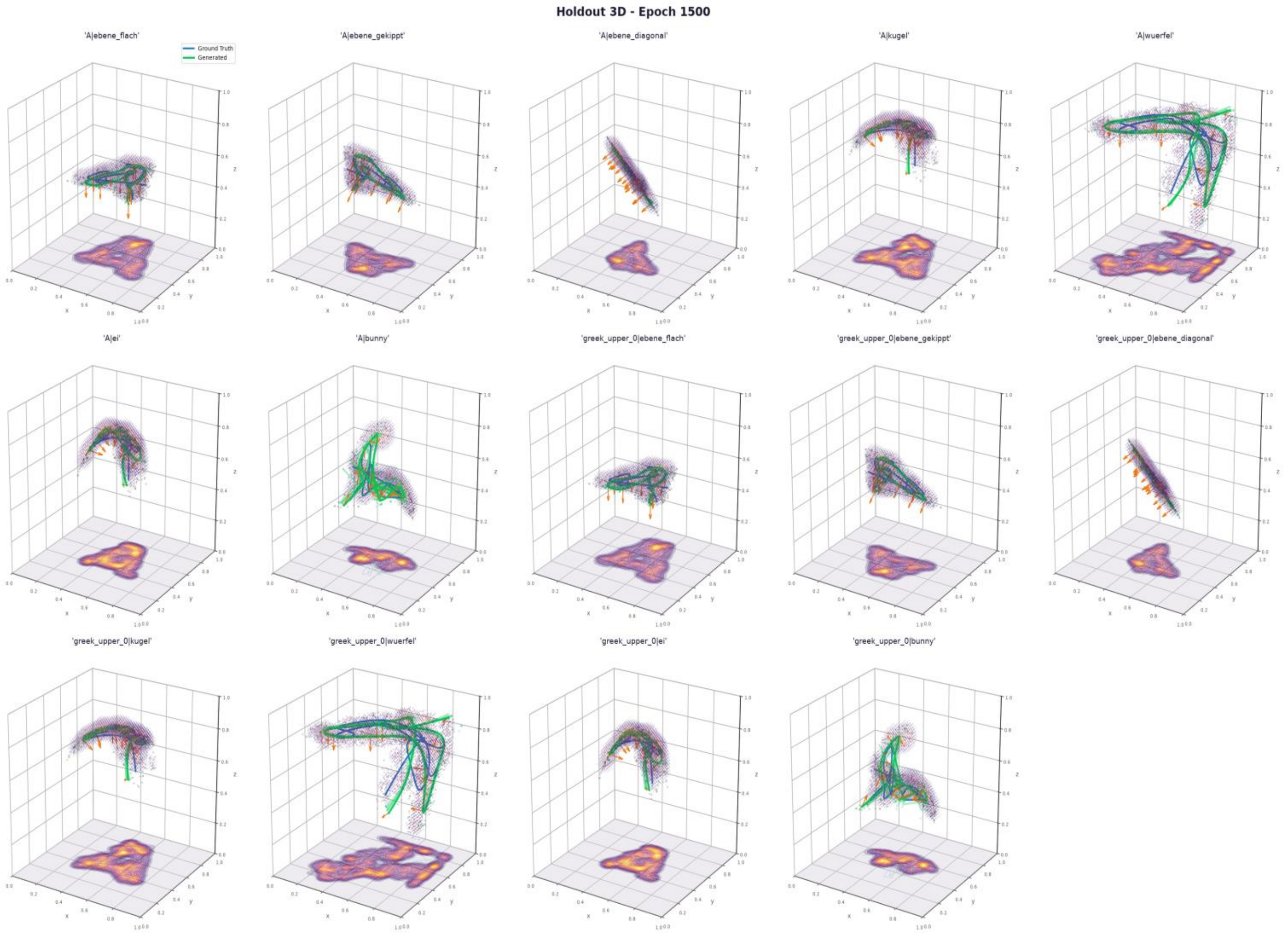
3D transfer

- First step:
  - Add third dimension to 2D database (just as 0)
  - Train same architecture to generate B-Spline cps with third dimension
  - Test generalizability to different plane orientations/shapes



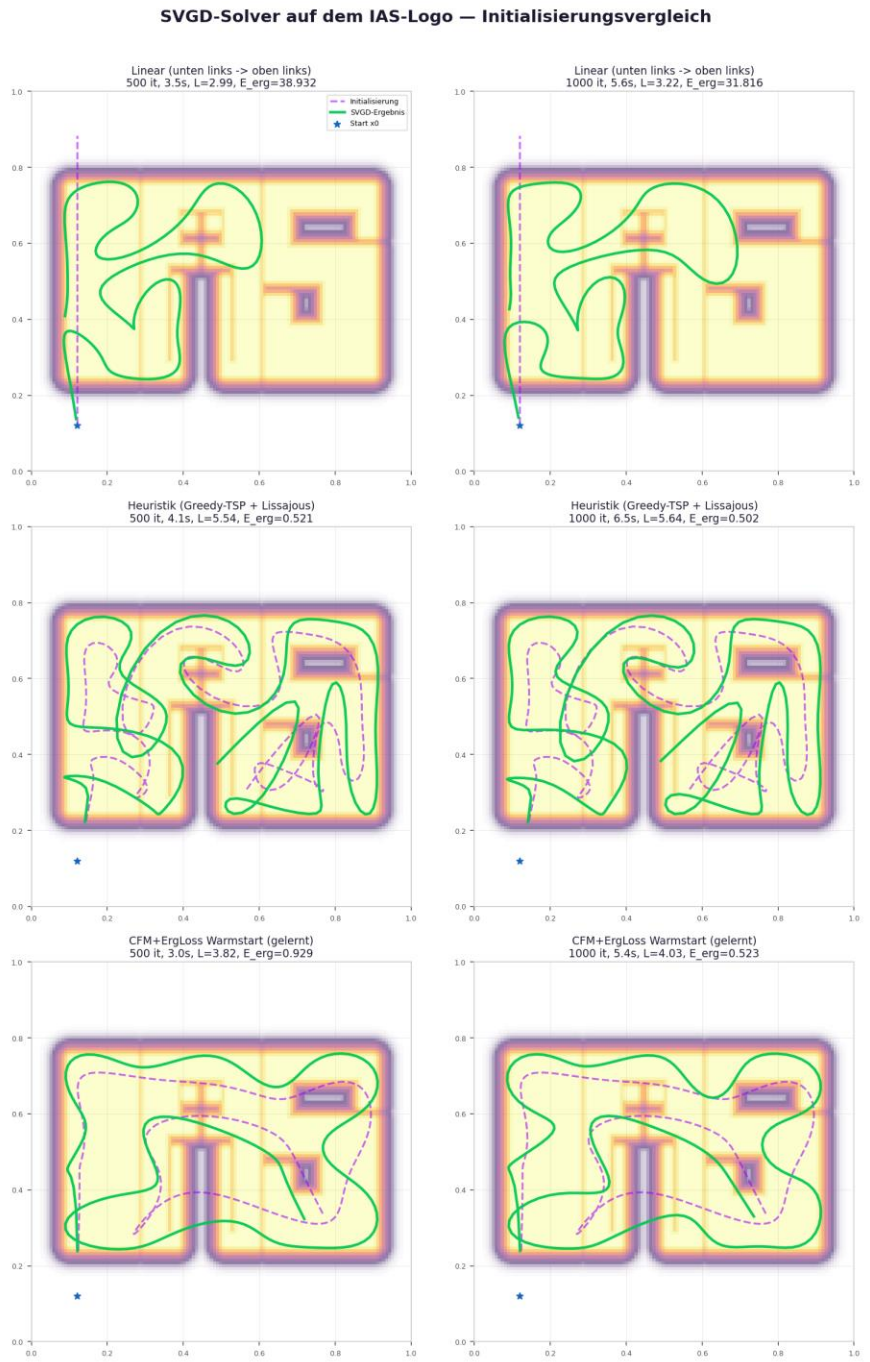
3D transfer

- second step:
  - Change database:
    - Project shapes and trajectories on different surfaces
    - Train on new database



First tests of comparison of initializations

	Ergodicity before svgd (1000 iterations)	Ergodicity after svgd (1000 iterations)
heuristic	1,628	0,502
cfm	1,015	0,523



Plan for next week

- Expand 3D database for more 3D shapes to learn generalization
  - Finetune existing models on these extension
- I´ntroduce more constraint…
- Make a larger analysis and comparison of different initializations (IAS logo)
- Try mujoco simulation
- Improve uncertainty exploration/exploitation
  - Make use of pathlength conditioning to limit exploration ➔ introduce a budget
  - Fair comparisons with similar pathlengths
  - Find and analyze a metric